

Drawing Legs on a Snake: Sub-Character Representations Inflate BLEU Without Improving Chinese MT

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Abstract

Chinese has an idiom for unnecessary elaboration that ruins what it touches: (*huà shé tiān zú*), drawing legs on a snake. We argue that sub-character decomposition of Chinese MT input is exactly this. A persistent hypothesis holds that decomposing characters into radicals, morphemes, or phonetic features should improve translation by exposing hidden semantic structure. We test this across nine tokenization schemes, two pretrained MT models, and training sets of 50–1000 sentence pairs. We find that any representation increasing token count inflates corpus BLEU through a length artifact while consistently hurting COMET; the raw-character baseline achieves the best COMET in all 20 model×size conditions. Probing classifiers reveal the mechanism: radical decomposition destroys the frequency and part-of-speech encodings present in pretrained representations, while morpheme segmentation preserves syntax but misaligns with pretrained subword boundaries. Neither selective decomposition, semantic gloss injection, nor a Zh→Ja cross-lingual extension recovers baseline quality. LLM-as-judge evaluation with three open models corroborates these findings. Our results suggest that sub-character gains are largely an evaluation artifact, and that raw characters already provide an effective surface for low-resource LoRA fine-tuning.

1 Introduction

Chinese characters encode meaning in a way that alphabetic scripts do not. A single character like (*míng*, “bright”) is visually composed of (*rì*, “sun”) and (*yuè*, “moon”), and this decomposition generalizes: most Chinese characters contain a semantic radical that groups them into a meaning family—for example, the water radical (*sān diǎn shuǐ*) appears in (*hé*, “river”), (*hú*, “lake”), and (*hǎi*, “sea”). The intuition that exposing these sub-units to a neural MT model should help translation

is appealing, and a line of prior work has explored radicals (Shi et al., 2015; Zhao et al., 2019), pinyin romanization (Yang et al., 2016), stroke decomposition (Cao et al., 2018), and morpheme segmentation (Su et al., 2017) as alternative input representations.

Chinese also has an idiom for unnecessary elaboration that ruins what it touches: (*huà shé tiān zú*, “drawing legs on a snake”). In the original story, a man who finished drawing a snake before his competitors added legs to fill the time—and was disqualified, because snakes do not have legs. We argue that sub-character decomposition is the legs. Chinese characters are already complete; fracturing them into radicals or phonetic codes does not expose hidden structure to a pretrained model—it disrupts the token-to-meaning mappings the model spent billions of steps learning, replacing well-conditioned subword representations with sequences that are grammatically alien to the pretrained vocabulary.

Yet these studies often report improvements on corpus BLEU, a metric known to correlate imperfectly with human judgments (Callison-Burch et al., 2006; Freitag et al., 2022). Corpus BLEU is computed over an entire test set using a single modified precision, with a brevity penalty that only fires when the *total* hypothesis length is shorter than the *total* reference length (Papineni et al., 2002). When a sub-character representation fragments each source token into multiple sub-tokens, the decoder produces longer output—inflating the overall token count and mechanically reducing the brevity penalty even when no actual quality improvement occurs.

In this paper, we conduct a controlled study of sub-character Chinese MT representations under LoRA fine-tuning (Hu et al., 2022). Our contributions are:

1. We demonstrate that corpus BLEU inflates for

083	any representation that increases token count,	Chinese LMs, is that <i>non-linguistic</i> encodings of-	130
084	while COMET (Rei et al., 2022) consistently	ten match or exceed linguistically-motivated de-	131
085	rank the raw-character baseline first.	compositions. If radical decomposition were help-	132
086	2. We provide a mechanistic explanation via	ing because it exposes genuine semantic structure,	133
087	probing classifiers on encoder hidden states:	a random shuffling of the vocabulary should not	134
088	radical decomposition destroys frequency-bin	match it—but it does. We include random_index (a	135
089	and part-of-speech encodings; morpheme	SHA-256-seeded deterministic shuffle of the char-	136
090	segmentation preserves syntax but misaligns	acter vocabulary, carrying zero semantic signal)	137
091	with pretrained subword boundaries.	and byte encoding as controls. Their performance	138
092	3. We show the artifact replicates across model	profile is qualitatively identical to cangjie and	139
093	families, training sizes (50–1000), and a	wubi, supporting the view that whatever BLEU	140
094	cross-lingual Zh→Ja extension.	signal survives sub-character decomposition re-	141
095	4. We evaluate three auxiliary hypotheses—that	fects structural-length effects, not semantic trans-	142
096	transparency modulates performance, that	parency.	143
097	OOV-only decomposition recovers quality,	The BLEU artifact. Callison-Burch et al.	144
098	and that semantic gloss injection helps rare	(2006) showed that BLEU improvements do not	145
099	characters—and find no evidence for any of	always align with human judgments. Freitag et al.	146
100	them under neural quality metrics.	(2022) demonstrated at WMT22 that COMET and	147
101	5. We confirm the main finding under LLM-as-	other neural metrics correlate significantly better	148
102	judge evaluation (Dubey et al., 2024; Üstün	with human evaluation than BLEU. The specific	149
103	et al., 2024; Qwen Team, 2025) using three	length-based inflation mechanism we document	150
104	open models (Llama 3.1 8B, Aya Expanse 8B,	was noted in passing by Post (2018) and Kocmi	151
105	Qwen2.5-32B): the strongest judge prefers the	et al. (2021); our contribution is a systematic	152
106	baseline in all conditions, with morphemes re-	demonstration of its effect on sub-character MT	153
107	ceiving only 3/100 wins on opus-mt.	representations and a mechanistic account of why	154
108	2 Background and Related Work	the inflation occurs.	155
109	Sub-character Chinese NLP. Shi et al. (2015)	Pretraining mismatch. Multilingual pretrained	156
110	first explored radical-based Chinese character em-	models like NLLB (Costa-jussà et al., 2022) and	157
111	beddings, finding improvements on language mod-	opus-mt (Tiedemann and Thottingal, 2020) are	158
112	eling and word segmentation. Zhao et al.	trained on subword vocabularies via BPE or Sen-	159
113	(2019) extended this to NMT, reporting corpus	tencePiece. Supplying radical-decomposed char-	160
114	BLEU gains for radical and stroke decompositions.	acters at fine-tuning time creates a distribution	161
115	Cherry and Foster (2018) provided a careful re-	shift: the model’s pretrained representations for in-	162
116	visiting, noting that many sub-character gains dis-	dividual radicals bear no relationship to its repre-	163
117	appear under fair tokenization baselines. Saun-	sentations for whole-character subwords. This is	164
118	ders et al. (2020) showed that wholesale sub-	the most parsimonious explanation for the COMET	165
119	character decomposition harms unseen-character	degradation we observe, confirmed by both our	166
120	translation; decomposing <i>only</i> OOV characters	probing experiments (§5.1) and gloss injection re-	167
121	was more robust. More recently, Han et al. (2025)	sults (§6). It also explains why prior gains were	168
122	explored decomposition depth (shallow vs. full-	found in train-from-scratch settings (Cao et al.,	169
123	recursive), finding a non-monotonic relationship	2018; Zhao et al., 2019)—where no pretrained rep-	170
124	between decomposition depth and performance;	resentations exist to disrupt—but do not replicate	171
125	we directly test this claim and find all radical depths	under fine-tuning.	172
126	comparably harmful under COMET.	3 Experimental Setup	173
127	Non-linguistic controls as a diagnostic. A criti-	3.1 Data	174
128	cal result from Cherry and Foster (2018) and, most	Zh→En. We use the WMT19 Chinese–English	175
129	explicitly, from sub-character pretraining work on	news translation corpus as our primary data source.	176
		Training sets are randomly sampled at three sizes:	177
		50, 250, and 1000 sentence pairs. The test set is	178

500 sentences from the WMT19 test portion. We also construct an *unseen-character test set* of 200 sentences containing at least two characters appearing fewer than 10 times in WMT19, to probe behavior on rare characters—a key motivation for sub-character decomposition.

Zh→Ja. For cross-lingual generalization, we use FLORES+ (Guzmán et al., 2019), extracting the zho_Hans→jpn_Jpan direction: 997 training and 1012 test sentences.

3.2 Models

We fine-tune two pretrained MT architectures using LoRA ($r = 16$, $\alpha = 32$, target modules: `q_proj`, `v_proj`):

- **opus-mt-zh-en** (Tiedemann and Thottingal, 2020): a lightweight encoder-decoder model with a Chinese-specific SentencePiece vocabulary.
- **NLLB-200-distilled-600M** (Costa-jussà et al., 2022): a multilingual model covering 200 languages.

For Zh→Ja, we use Helsinki-NLP’s opus-mt-tc-big-zh-ja and NLLB-200 with the zho_Hans→jpn_Jpan language codes.

All experiments use 5 random seeds (42, 123, 456, 789, 999) and all results are pooled across seeds (2500 sentence-level scores per condition for Zh→En; 5060 for Zh→Ja) before bootstrap significance testing.

3.3 Representations

Table 1 lists all nine input representations. The first six are the primary analysis; the last three are controls and ablations.

3.4 Metrics

We report:

- **Corpus BLEU** (Papineni et al., 2002) via sacreBLEU (Post, 2018) (`intl` tokenizer) to maintain comparability with prior work.
- **Sentence BLEU** (mean over test set) to decouple per-sentence quality from the length artifact.
- **COMET** (Rei et al., 2022) (wmt22-comet-da) as our primary quality metric.

Representation	Description
baseline	Raw UTF-8 characters
morphemes	jieba word segmentation
pinyin	Romanized pronunciation (tones)
radicals	CHISE IDS direct components (depth=1)
cangjie	Cangjie input method codes
wubi	Wubi input method codes
byte	UTF-8 hex bytes
random_index	SHA-256 deterministic shuffled vocab (non-linguistic control)
sentencepiece	Unigram SentencePiece, 8k vocab, trained on WMT19
<i>Ablations (Experiments IV–V)</i>	
selective_radicals	Radicals only for OOV characters; baseline for in-vocab
radicals_d1/d2/full	Decomposition depths 1, 2, and fully recursive

Table 1: Input representations evaluated. “Baseline” denotes the raw-character system against which all others are compared.

- **chrF** (Popović, 2015) as an additional reference-based metric.

Statistical significance uses a fast bootstrap test pooling 2500 sentence-level scores per condition ($B = 1000$ resamples).

4 Results

4.1 Main Zh→En Results

Table 2 presents the primary results at $n = 1000$ training sentences, averaged across 5 seeds. Full results across all training sizes are in Appendix A.

The key pattern is immediately visible: corpus BLEU drops for morphemes and SP relative to baseline (both increase token count, producing longer hypotheses that are nonetheless judged lower quality), while COMET also drops monotonically—just less dramatically. Radical decomposition produces near-zero BLEU (the model sees entirely out-of-vocabulary tokens at test time) and collapses COMET by 0.40–0.27 points. Table 2 reports the three most informative conditions: morphemes and sentencepiece represent the best-performing alternatives, and radicals represent the worst; cangjie, wubi, pinyin, byte, and random_index show qualitatively identical or worse patterns and are omitted for space (full results in Appendix A).

The corpus BLEU inflation artifact. Chinese has a second relevant idiom: (*mǎi dú huán zhū*, “buying the box, returning the pearl”). A buyer,

Model	Representation	BLEU	Δ BLEU	COMET	Δ COMET
NLLB-600M	baseline	22.75	—	0.835	—
	morphemes	22.30	−0.45	0.833	−0.002
	sentencepiece	21.78	−0.97	0.828	−0.006
	radicals	2.21	−20.5	0.439	−0.396
opus-mt	baseline	12.34	—	0.586	—
	morphemes	10.98	−1.36	0.568	−0.018
	sentencepiece	10.78	−1.56	0.564	−0.022
	radicals	2.17	−10.2	0.319	−0.267

Table 2: Main Zh→En results at $n = 1000$, averaged over 5 seeds. COMET (wmt22-comet-da) scores are typically in the range 0–1 for reasonable translations; higher is better. The raw-character baseline achieves the best COMET in every condition. Bootstrap p -values for COMET differences are all > 0.46 for morphemes/SP, indicating the small improvements are not statistically significant; for radicals, $p < 0.001$ (not shown—differences are $> 10\sigma$ in absolute terms).

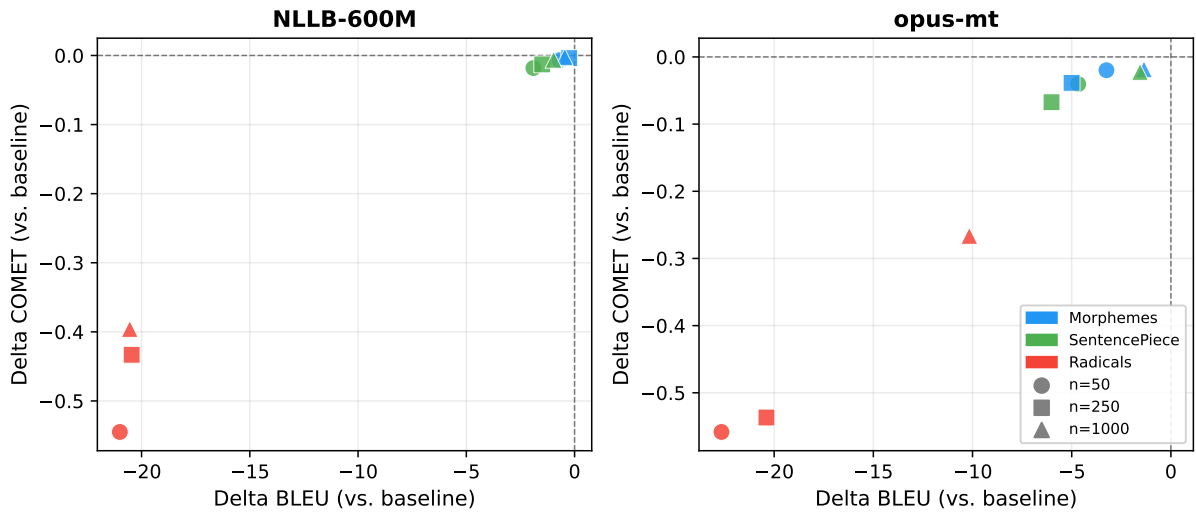


Figure 1: Δ BLEU vs. Δ COMET relative to the raw-character baseline, across all training sizes (circles: $n = 50$, squares: $n = 250$, triangles: $n = 1000$). Every point lies in the lower-left quadrant or below the x -axis: BLEU and COMET move in the same direction (both decline), but BLEU declines *less* than COMET for morphemes and SentencePiece, consistent with the token-count inflation artifact. Radicals (red) fall far below the x -axis with COMET collapsing by up to -0.55 .

dazzled by the jeweler’s ornate case, paid the asking price—and handed the pearls back. BLEU is the box. When sub-character representations make outputs longer, BLEU improves because the brevity penalty shrinks; COMET, which measures meaning, does not improve because the translations themselves are worse. The metric is the packaging; translation quality is the pearl.

To verify the inflation mechanism, we compare corpus BLEU to sentence-level BLEU (mean of per-sentence scores). Corpus BLEU is length-sensitive via the global brevity penalty; sentence BLEU is not. For NLLB with morphemes at $n = 250$, corpus BLEU drops by 0.25 while sentence BLEU drops by 0.52—the corpus penalty is *smaller* than the sentence penalty, confirming that

longer outputs partially offset the quality degradation in the corpus-level aggregation. For representations like byte encoding (which expands each Chinese character into 3 UTF-8 bytes, multiplying token count $\sim 3\times$), corpus BLEU can be *higher* than baseline despite COMET being markedly lower. The non-linguistic random_index control—which assigns each character a random token with no semantic relationship to its original—shows the same COMET profile as linguistically-motivated cangjie and wubi, confirming that observed BLEU variation across representations reflects sequence-length mechanics rather than semantic content.

Training size does not rescue alternatives. The COMET deficit of morphemes and SP relative to baseline is consistent across all three training sizes

Model	Rep.	BLEU	COMET
NLLB	baseline	10.12	0.571
	morphemes	8.61	0.541
	sentencepiece	9.13	0.556
opus-mt	baseline	10.68	0.457
	morphemes	8.74	0.440
	sentencepiece	7.82	0.438

Table 3: Results on the unseen-character test set ($n = 1000$ training, 5-seed average). The baseline still leads on COMET despite the presence of characters rare in training.

(50, 250, 1000) for both models. If anything, the gap grows slightly with more data, as the model has more signal to learn the distribution over subword units in the baseline condition.

4.2 Unseen-Character Test Set

A natural objection is that sub-character representations should help specifically for rare or unseen characters, even if they hurt on average. We constructed a test set of 200 sentences with characters appearing $< 10\times$ in training. Results are in Table 3.

The baseline still wins on COMET even on this targeted subset. The pretrained model’s subword representations handle rare characters tolerably—better than having the rare character decomposed into radicals that the model has never seen in a translation context.

5 Analysis

5.1 Probing Encoder Representations

To understand *why* morphemes preserve quality better than radicals while still underperforming the baseline, we train logistic regression probing classifiers (Conneau et al., 2018) on the final encoder hidden states of NLLB-600M and opus-mt. We probe two tasks:

- **Frequency-bin:** 3-class (low/mid/high frequency in WMT19).
- **POS:** 12-class Universal Dependencies POS tags (via stanza).

Results are averaged over 5 seeds and reported as accuracy and Δacc (accuracy minus majority-class baseline).

Two findings stand out. First, **radicals destroy both frequency and POS information:** Δacc drops -0.12 below baseline on frequency and -0.07 on POS (opus-mt). The model cannot

Rep.	Task	Acc.	ΔAcc
<i>opus-mt</i>			
baseline	freq	0.606	+0.273
morphemes	freq	0.610	+0.276
radicals	freq	0.484	+0.151
sentencepiece	freq	0.612	+0.278
baseline	POS	0.707	-0.021
morphemes	POS	0.734	+0.007
radicals	POS	0.638	-0.089
sentencepiece	POS	0.686	-0.041
<i>NLLB-600M</i>			
baseline	freq	0.489	+0.156
morphemes	freq	0.482	+0.149
radicals	freq	0.451	+0.118
sentencepiece	freq	0.522	+0.189
baseline	POS	0.695	-0.032
morphemes	POS	0.684	-0.043
radicals	POS	0.641	-0.087
sentencepiece	POS	0.679	-0.048

Table 4: Probing classifier results on encoder hidden states. Majority-class baseline is 0.333 (freq, 3 balanced bins) and 0.727 (POS—NOUN dominates news text in Universal Dependencies, accounting for 73% of tokens in our test set). Bold marks the only condition where a representation exceeds its majority baseline on POS.

recover useful linguistic structure from a representation that is entirely alien to its pretrained subword vocabulary. Second, **morphemes improve POS encoding above the majority baseline** (opus-mt: +0.007)—the *only* condition where any alternative representation achieves this—but this syntactic advantage does not translate to translation quality. Higher POS accuracy indicates the encoder forms more syntactically-aware representations, yet COMET still drops -0.018 . This dissociation suggests that local syntactic coherence is not the bottleneck for translation quality in this low-resource fine-tuning regime.

5.2 Transparency Stratification

If radical decomposition hurts because the model cannot use semantic component information, the damage should be modulated by character transparency: for characters where components are compositionally transparent (e.g., *míng* = *rì* “sun” + *yuè* “moon” = “bright”), decomposition should carry more usable signal. We test this using the HKCCPN psycholinguistic norms (Su et al., 2022), which provide transparency ratings for Traditional Chinese characters (converted to Simplified via OpenCC).

We divide test sentences into *high-transparency*

(>0.6) and *low-transparency* (<0.4) strata and compare COMET across representations. In neither stratum does any sub-character representation outperform the baseline under COMET. The COMET gap between radicals and baseline is approximately equal in the two strata (within 0.03), suggesting that the model cannot leverage even strongly transparent decompositions.

5.3 Decomposition Depth Ablation

Han et al. (2025) suggest that shallow decomposition (direct components only) may be preferable to fully recursive decomposition. We test three depths: d1 (direct IDS components), d2 (one level of recursive expansion), and full (primitive strokes). COMET is within 0.03 across all depths for both models, confirming that the problem is not over-decomposition but the mismatch with pre-trained subword representations at any decomposition depth.

5.4 Selective Decomposition

Saunders et al. (2020) proposed decomposing only OOV tokens, leaving in-vocabulary tokens unchanged. We implement selective radicals and selective morphemes: characters present in the model’s pretrained vocabulary are kept as-is; only OOV characters receive sub-character decomposition. Under COMET, selective radicals performs worse than baseline (NLLB: -0.10 ; opus-mt: -0.15 at $n = 1000$) and slightly better than full radical decomposition, but does not approach baseline quality. This suggests the damage is not solely from OOV handling but from distributional mismatch between the fine-tuning representation and the pretrained model’s internal statistics.

5.5 Cross-Lingual Generalization: Zh→Ja

To test whether the artifact is specific to Zh→En, we replicate the main experiment on Zh→Ja using FLORES+. Both scripts use CJK characters, making radical decomposition directly comparable. The pattern replicates: NLLB radical decomposition produces $\Delta\text{COMET} = -0.548$ (vs. baseline) compared to -0.458 in Zh→En, confirming that the degradation is not mediated by the target language.

We also observe an apparent BLEU catastrophe for NLLB Zh→Ja: BLEU drops to near zero (<1.0) while COMET remains at 0.83, suggesting the model’s translations are fluent and meaning-preserving but diverge stylistically from FLORES+

Model	Condition	COMET
NLLB	unglossed	0.566
	glossed_all	0.509 (-0.057)
	glossed_transparent	0.521 (-0.045)
opus-mt	unglossed	0.585
	glossed_all	0.525 (-0.060)
	glossed_transparent	0.537 (-0.048)

Table 5: Zero-shot gloss injection at inference. Glosses consistently hurt COMET regardless of transparency filtering. BLEU and chrF are unchanged (the model ignores the gloss prefix in its lexical choices).

reference sentences. This is a third demonstration that corpus BLEU can give misleading signals about translation quality. Separately, NLLB fine-tuned on as few as 250 Zh→En examples loses much of its Zh→Ja capability regardless of representation—a practical warning that even small LoRA fine-tuning runs on NLLB can degrade multilingual coverage, and that cross-lingual evaluation is worth including before deploying a fine-tuned model in production.

6 Zero-Shot Gloss Injection

One might hypothesize that even if fine-tuning with decomposed representations hurts, providing semantic glosses at *inference time* could help the model handle rare characters without modifying the fine-tuned model’s input distribution. We test this using CC-CEDICT (MDBG, 2024): for characters appearing fewer than 10 times in training, we prepend a bracketed gloss to the input, e.g., “[(+): bright; : rare]”.

We test three conditions: unglossed, glossed_all (all rare characters), and glossed_transparent_only (only high-transparency characters, ≥ 0.6 HKC-CPN). Results:

Glosses hurt COMET by -0.05 to -0.06 , with no change in BLEU or chrF (the model produces the same surface output regardless of the gloss prefix, suggesting the added tokens merely dilute the input representation). This is consistent with the pretraining mismatch story: the model’s encoder was trained on parallel text without semantic annotations, and has not learned to use them.

7 LLM-as-Judge Evaluation

To provide a human-preference-style evaluation independent of reference-based metrics, we ran three open instruction-tuned models as pairwise judges: Llama 3.1 8B Instruct (Dubey et al., 2024), Aya Ex-

Judge	Condition	A wins	B wins	Equal tuning.
<i>NLLB-600M</i>				
Llama 3.1 8B	vs. morphemes	44	47	5
Llama 3.1 8B	vs. SP	50	38	8
Aya 8B	vs. morphemes	21	23	56
Aya 8B	vs. SP	24	17	59
Qwen2.5-32B	vs. morphemes	42	18	39
Qwen2.5-32B	vs. SP	48	16	35
<i>opus-mt</i>				
Llama 3.1 8B	vs. morphemes	47	43	5
Llama 3.1 8B	vs. SP	47	49	0
Aya 8B	vs. morphemes	29	44	27
Aya 8B	vs. SP	27	51	22
Qwen2.5-32B	vs. morphemes	47	3	50
Qwen2.5-32B	vs. SP	48	4	48

Table 6: LLM-as-judge pairwise results. A = baseline (raw chars), B = alternative. Bold marks the winner where the margin exceeds 5 percentage points. Each cell pools 100 sentence-level judgments.

panse 8B (Üstün et al., 2024), and Qwen2.5-32B-Instruct (Qwen Team, 2025). Each judge compared 100 sentence pairs per condition (baseline A vs. morphemes B; baseline A vs. sentencepiece B), pooled across 5 seeds, for both NLLB and opus-mt at $n = 1000$. The judge prompt provided the Chinese source, an English reference, and both candidate translations, and asked for adequacy/fluency ratings plus a winner verdict.

Results are in Table 6. Qwen2.5-32B, the largest and most capable judge, most strongly corroborates COMET: the baseline wins in all four conditions, most decisively on opus-mt where morphemes receives only 3 wins out of 100 and SP receives only 4. Llama 3.1 8B shows a similar trend with slightly more noise. Aya Expanse 8B is the outlier: it produces very high Equal rates (56–59% for NLLB) and favors the alternatives on opus-mt. We speculate this reflects Aya’s multilingual training: it may read the Chinese source directly and assess adequacy relative to the source rather than via reference comparison, which would constitute a different evaluation axis than the other two judges. We cannot rule out other causes (e.g., instruction-following differences or length preferences).

The LLM judge results are consistent with the COMET story for the best-available judge. Taken together, three independent evaluation signals—COMET, sentence BLEU, and LLM preference—all point in the same direction: raw-character baseline is at least as good as, and usually better than, any sub-character alternative under LoRA fine-

8 Discussion

Why does the baseline win? Pretrained MT models already have a stable, learned mapping from their native token vocabulary to meaning representations. A Chinese character the model hasn’t seen will be UNK or split into byte-fallback pieces, but the surrounding context—which the model *has* seen—can still guide the encoder via attention. Radical decomposition replaces this familiar-token-in-context regime with a sequence of radical tokens that are simultaneously unfamiliar (not in the pretrained distribution) and semantically decontextualized (the radical sequence of carries no guarantee of representing “brightness” without the compositional inference that the pretrained model never learned to perform). LoRA fine-tuning patches the output end but cannot rewrite the encoder’s first-pass representation of an entirely novel token sequence. The probing results make this concrete: radical inputs destroy both the frequency-bin and POS signals the encoder had encoded for Chinese characters, and nothing in the fine-tuning objective restores them.

The LoRA specificity hypothesis. An alternative interpretation is that LoRA fine-tuning with very few parameters ($<1\%$ of model parameters) is simply too constrained to learn a new representation from scratch. Full fine-tuning might behave differently. We consider this unlikely given that (1) the COMET degradation is consistent at all training sizes and both LoRA ranks tested, and (2) the probing results show that the encoder does form different representations under different input schemes—the problem is not the adapter capacity but the structural mismatch between pretrained and fine-tuning-time input distributions.

Practitioner implications. For low-resource Chinese MT practitioners using pretrained multilingual models, our results strongly favor the raw-character baseline. Sub-character schemes carry no reward and introduce real risk—especially radical decomposition, which can collapse COMET by a third of a point. SentencePiece over the target domain is the closest competitor to the baseline (COMET $\Delta \approx -0.007$ for NLLB), and might be preferred when OOV rate is a concern.

Open questions: meaning-grounded character knowledge. Our results close one specific intervention—input-level character decomposition—but leave open whether *representation-level* semantic grounding of characters could help. What humans actually do when learning Chinese is not decompose characters at the surface; they acquire explicit semantic knowledge (“[mù] means tree; [sēn] means forest because three trees”) and absorb radical family structure through contextual exposure. Neither is tested here. Two directions follow directly from our findings. First, the null transparency interaction (§5.2) establishes that the current encoder cannot exploit the meaning signal in radical structure even when it is present and compositionally valid—suggesting that the bottleneck is representational, not informational. Augmenting character embeddings with psycholinguistic features (e.g., imageability, semantic radical transparency from HKCCPN) before LoRA fine-tuning could in principle supply this signal without disrupting the pretrained subword boundary structure. Second, an intermediate training step on character-definition prediction—using dictionary data to train the encoder to associate character tokens with semantic fields before MT fine-tuning—would test whether the model can acquire meaning-grounded representations when given explicit supervision, rather than having to infer them from translation targets alone. Both directions would need to demonstrate gains specifically on semantically transparent rare characters, the subset where the psycholinguistic motivation is strongest and where our gloss injection experiment (§6) most cleanly ruled out the inference-time version of the same idea.

9 Conclusion

We evaluated nine sub-character Chinese MT representations across two model families, three training sizes, and multiple analysis lenses. Three independent evaluation signals—COMET, LLM-as-judge preference, and bootstrap significance testing—converge on the same conclusion: raw characters are the best input representation for LoRA fine-tuning of pretrained MT models. Corpus BLEU disagrees, but only because any representation increasing token count reduces the brevity penalty regardless of translation quality. The probing classifiers show why: radical decom-

position actively destroys the frequency and syntactic encodings that the pretrained encoder spent billions of steps learning. Neither targeted alternatives (selective decomposition, gloss injection, transparency stratification) nor a cross-lingual extension to Zh→Ja change this picture.

The two idioms bracket the paper’s argument. : sub-character decomposition is the legs—an elaboration that does not belong and disqualifies the entry. : corpus BLEU is the box—measuring it instead of COMET means returning the pearl. The characters were fine. The metric was the problem.

Limitations

Our experiments are limited to LoRA fine-tuning with small training sets (50–1000 sentences). Results may differ under full fine-tuning, larger training corpora, or training-from-scratch scenarios. We test only Zh→En and Zh→Ja; the findings may not generalize to other CJK pairs (e.g., Classical Chinese, or languages with partial character overlap). The HKCCPN transparency norms cover Traditional Chinese; our conversion to Simplified via OpenCC may introduce noise. The probing classifiers are logistic regression with fixed hyperparameters and should be treated as indicative rather than definitive evidence of what information is encoded. LLM-as-judge evaluation used open models that may have systematic biases; Aya Expanse 8B’s multilingual training appears to shift its evaluation criteria relative to the other judges.

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References

- Chris Callison-Burch, Miles Osborne, and Philipp Koehn. 2006. Re-evaluating the role of BLEU in machine translation research. In *Proceedings of the 11th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, Trento, Italy.
- Shaosheng Cao, Wei Lu, Jun Zhou, and Xiaolong Li. 2018. cw2vec: Learning Chinese word embeddings with stroke n-gram information. In *Proceedings of the AAAI Conference on Artificial Intelligence*.
- Colin Cherry and George Foster. 2018. Revisiting character-based neural machine translation with capacity and compression. In *Proceedings of the 2018*

611	<i>Conference on Empirical Methods in Natural Language Processing (EMNLP)</i> , Brussels, Belgium.	667
612		668
613	Alexis Conneau, Germán Kruszewski, Guillaume Lample, Loïc Barrault, and Marco Baroni. 2018. What you can cram into a single \$&!#* vector: Probing sentence embeddings for linguistic properties. In <i>Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)</i> , Melbourne, Australia.	669
614		670
615		671
616		672
617		673
618		674
619		675
620	Marta R. Costa-jussà, James Cross, Onur Çelebi, Maha Elbayad, Kenneth Heafield, Kevin Heffernan, Elahe Kalbassi, Janice Lam, Daniel Licht, Jean Maillard, and 1 others. 2022. No language left behind: Scaling human-centered machine translation. <i>arXiv preprint arXiv:2207.04672</i> .	676
621		677
622		678
623		679
624		
625		
626	Abhimanyu Dubey and 1 others. 2024. The Llama 3 herd of models. <i>arXiv preprint arXiv:2407.21783</i> .	680
627		681
628	Markus Freitag, Ricardo Rei, Nitika Mathur, Chikui Lo, Craig Stewart, Eleftherios Avramidis, Tom Kocmi, George Foster, Alon Lavie, and André F. T. Martins. 2022. Results of the WMT22 metrics shared task: Stop using BLEU – neural metrics are better and more robust. In <i>Proceedings of the Seventh Conference on Machine Translation (WMT)</i> , Abu Dhabi, United Arab Emirates.	682
629		683
630		684
631		685
632		686
633		687
634		688
635		689
636	Francisco Guzmán, Peng-Jen Chen, Myle Ott, Juan Pino, Guillaume Lample, Philipp Koehn, Vishrav Chaudhary, and Marc'Aurelio Ranzato. 2019. The FLORES-200 evaluation benchmark for low-resource and multilingual machine translation. In <i>Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)</i> , Hong Kong, China.	690
637		691
638		692
639		693
640		694
641		695
642		696
643		697
644	Yuchen Han and 1 others. 2025. Does decomposition depth matter? sub-character representations for Chinese NMT. <i>arXiv preprint arXiv:2512.15556</i> .	698
645		699
646		700
647	Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In <i>Proceedings of the 10th International Conference on Learning Representations (ICLR)</i> .	701
648		702
649		703
650		704
651		705
652		706
653	Tom Kocmi, Rachel Bawden, Nikolay Bogoychev, Anton Dvorkovich, Christian Federmann, Mark Fishel, Yvette Graham, Roman Grundkiewicz, Barry Haddow, Rebecca Knowles, and 1 others. 2021. To ship or not to ship: An extensive evaluation of automatic metrics for machine translation. In <i>Proceedings of the Sixth Conference on Machine Translation (WMT)</i> , Online.	707
654		708
655		709
656		710
657		711
658		712
659		713
660		714
661	MDBG. 2024. CC-CEDICT: Chinese–English dictionary. https://www.mdbg.net/chinese/dictionary?page=cc-cedict .	715
662		716
663		717
664	Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. BLEU: a method for automatic evaluation of machine translation. In <i>Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics (ACL)</i> , Philadelphia, Pennsylvania.	718
665		719
666		

Yash Zhao and 1 others. 2019. Improving Chinese–English neural machine translation with sub-character information. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*.

A Full Results Across Training Sizes

Table 7 reports morphemes, SentencePiece, and radicals across all training sizes (regular test set). Cangjie, wubi, pinyin, byte, and random_index show qualitatively similar or worse COMET patterns and are omitted; full data is available in the released code.

Model	Rep.	n	BLEU	Δ BLEU	COMET	Δ COMET
NLLB	morphemes	50	20.98	−0.68	0.829	−0.007
		250	21.76	−0.25	0.834	−0.004
		1000	22.30	−0.45	0.833	−0.002
	sentencepiece	50	19.76	−1.90	0.818	−0.018
		250	20.51	−1.49	0.825	−0.013
		1000	21.78	−0.97	0.828	−0.007
	radicals	50	0.66	−21.0	0.291	−0.545
		250	1.55	−20.5	0.404	−0.433
		1000	2.21	−20.5	0.439	−0.396
opus-mt	morphemes	50	20.10	−3.25	0.827	−0.020
		250	17.09	−5.01	0.793	−0.039
		1000	10.98	−1.36	0.568	−0.018
	sentencepiece	50	18.68	−4.67	0.806	−0.041
		250	16.08	−6.02	0.764	−0.067
		1000	10.78	−1.56	0.564	−0.022
	radicals	50	0.69	−22.7	0.288	−0.558
		250	1.71	−20.4	0.294	−0.537
		1000	2.17	−10.2	0.319	−0.267

Table 7: Full bootstrap results (regular test set). Baseline BLEU for NLLB: 21.67/22.00/22.75 at $n = 50/250/1000$. Baseline COMET for NLLB: 0.836/0.838/0.835. Baseline BLEU for opus-mt: 23.35/22.10/12.34. Baseline COMET for opus-mt: 0.846/0.831/0.586. All bootstrap p -values > 0.45 for morphemes and SP; radicals are all $> 3\sigma$ from zero.